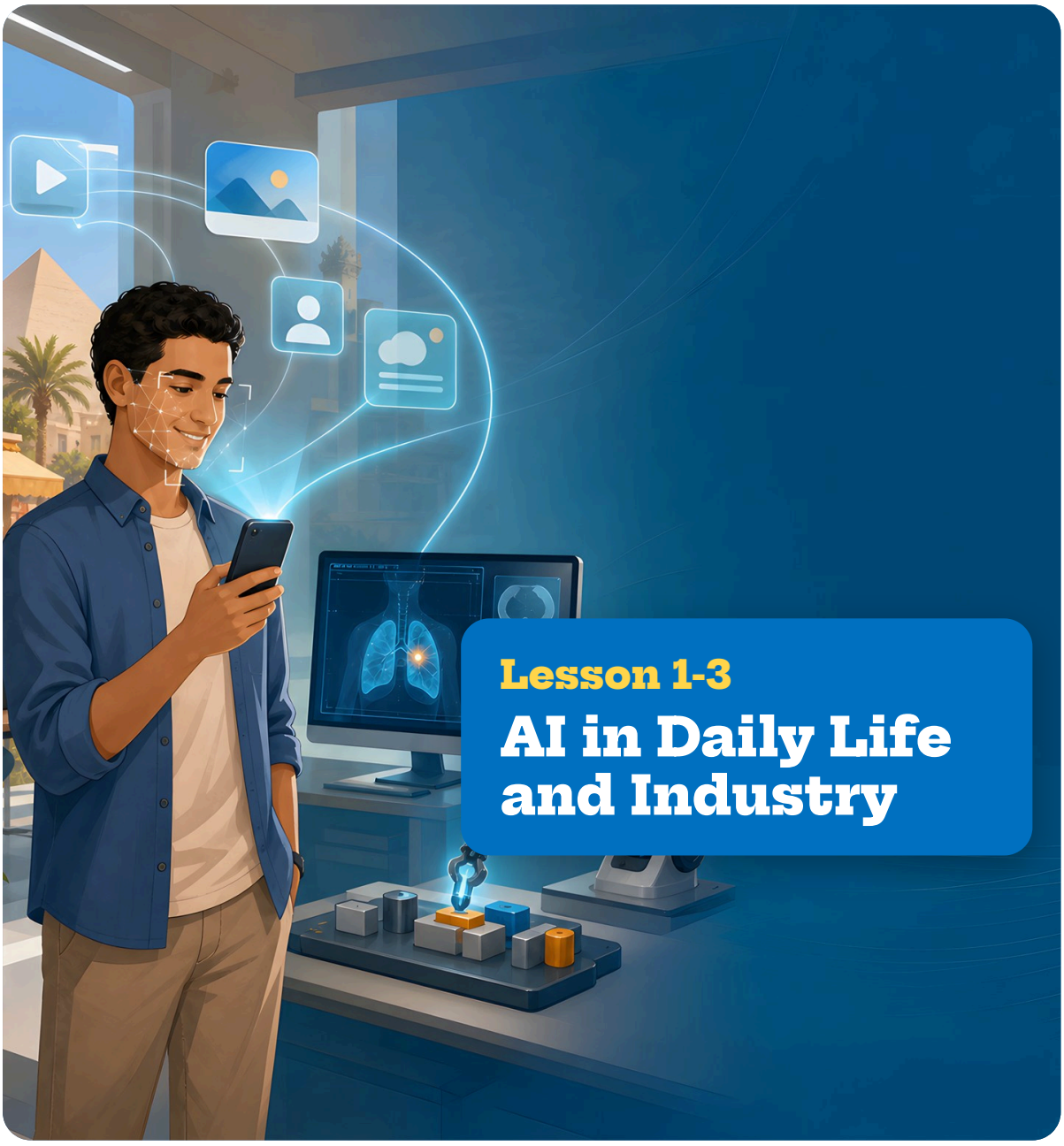


Learning Objectives

- Give examples of how AI is used in daily life and various industries.
- Explain what AI is good at and what requires caution when using AI.

AI is already part of many everyday services. A video site recommends the next clip, a voice assistant answers a spoken question, a translation tool converts text on a foreign menu, and a phone unlocks by recognizing your face. In industry, AI analyzes medical images, predicts when a machine will fail, and plans delivery routes. AI is effective at identifying patterns in data, but it is not suited to every task, and some uses require caution.

? Guiding question: Where is AI used in daily life and industry, what is AI good at, and what needs caution?

✎ Explore · In Pairs

With a partner, list three services you used this week that you think used AI (for example, a video recommendation, a voice assistant, or a translation). For each, predict what data the AI needed to do its job. Then decide together: which one would matter most if it made a mistake?

Key Fact:
AI is good at finding patterns in data, but human judgment is still needed where ethics, privacy, and responsibility are involved.

Key Concepts:

- recommendation system
- voice assistant
- machine translation
- face recognition
- image-diagnosis AI
- predictive maintenance
- black-box problem
- hallucination

📌 Point!

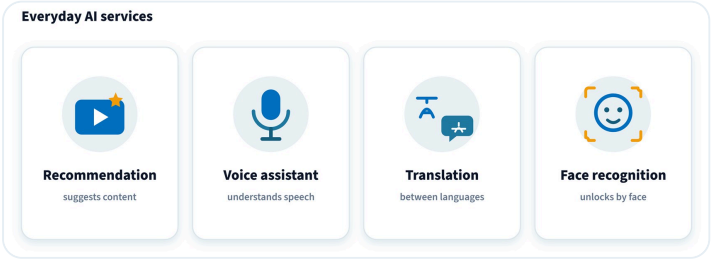
1. Use of AI in Daily Life

(1) Various AI technologies are already in use around us.

Service	Role of AI	Examples
Recommendation systems	Predict preferences from past behavior data and display recommendations.	YouTube, Amazon, Spotify, etc.
Voice assistants	Recognize a voice, understand commands, and execute them.	Siri, Google Assistant, etc.
Machine translation	Automatically translate text into different languages.	Google Translate, DeepL, etc.
Face recognition	Automatically detect and identify human faces in photographs.	Smartphone unlocking, etc.



Recommenders match new items to your past choices.



Four everyday-life AI services

KEY TERMS
recommendation system — a system that predicts preferences from past behavior data and displays recommendations.

2. Use of AI in Industry

(1) AI technologies are also being introduced in various industries.

Industry	Examples of AI use
Healthcare	Image-diagnosis AI (detecting diseases from X-ray and CT images), drug-discovery support
Agriculture	Predicting harvest timing, detecting pests and diseases
Manufacturing	Automation of product quality inspection, predictive maintenance
Logistics	Optimization of delivery routes



Four industries where AI is used

3. Characteristics of AI and Cautions for Use

(1) What AI is good at

- Finding and classifying patterns in complex data such as images and text
- Recognition and generation of images, audio, and text
- Probabilistic reasoning and prediction based on data

(2) What requires caution when using AI

- Ethical judgments (may lead to discrimination or prejudice)
- Judgments involving personal information or privacy
- Final decision-making (who is responsible for the result)
- Accuracy of judgments when training data is biased
- Generating content that differs from the facts (hallucination)
- Issues of rights when using copyrighted works as training data
- Loss of clarity about how a judgment was made (the black-box problem)

✎ Worked Example

(1) For each of a-d, match the AI technology used in the service with the most appropriate term from the choices below (A–D).

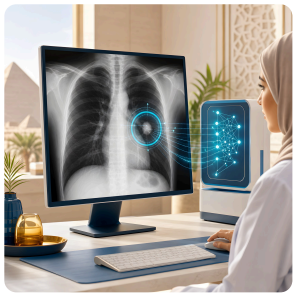
a YouTube recommends videos that match the user's preferences

b Speaking to a smartphone to check the weather

c Translating a foreign-language website into one's native language

d A smartphone camera automatically detects a person's face

[Choices] **A** Recommendation system **B** Voice assistant **C** Machine translation **D** Face recognition



In industry, AI flags what humans might miss.

💡 THINK IT THROUGH

A hospital is considering using image-diagnosis AI to detect diseases from X-ray images. Explain why the final diagnosis should be confirmed by a human doctor rather than left to AI alone.

💡 PAUSE & THINK

AI services often use your personal data — what you watch, buy, or say. Why does this raise privacy concerns, and who should decide how your data is used?

KEY TERMS

black-box problem — when it is unclear how an AI reached its judgment.

(2) Mark each statement (A–D) with ○ if true or × if false.

A AI is good at probabilistic reasoning and prediction based on data.

B AI is good at recognizing images and audio.

C It is fine to leave ethical judgments entirely to AI.

D AI judgments can become inaccurate when training data is biased.

✓ Solution

(1)

a: Recommendations matched to preferences come from a recommendation system. **A**

b: Operating by speaking is a voice assistant. **B**

c: Translating a foreign language into one's native language is machine translation. **C**

d: Detecting a person's face is face recognition. **D**

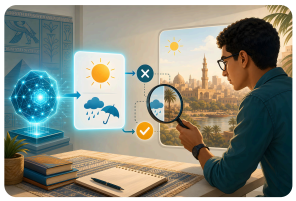
(2)

A AI is good at probabilistic reasoning and prediction based on data. Therefore, ○.

B AI is good at recognizing images, audio, and text. Therefore, ○.

C AI judgments can lead to discrimination or prejudice, so ethical judgments should not be left entirely to AI. Therefore, ×.

D Bias in training data affects the accuracy of AI judgments. Therefore, ○.



AI can be confidently wrong — verify its output.

Try

1 Answer the following questions.

1. What is the term for the system that predicts preferences from past behavior data and displays recommendations?
2. What is the term for the AI technology that recognizes a voice, understands commands, and executes them?
3. In which industry are image-diagnosis AIs used?
4. In manufacturing, what is the term for the system that predicts product failures in advance?

EXAM-STYLE QUESTION

Analyze why AI is well suited to some tasks but requires caution in others. **[6] In your answer:** refer to what AI is good at (finding patterns, recognition, prediction) and to what needs caution (ethical judgments, privacy, final decision-making).

2 Answer the following questions.

(1) From the following options (A–D), choose the one that does NOT belong to "what requires caution when using AI."

A Ethical judgments

B Finding and classifying patterns

C Judgments involving personal information or privacy

D Final decision-making

(2) For each AI usage example (a–d), indicate which industry is most closely related: **A** Healthcare, **B** Agriculture, **C** Manufacturing, **D** Logistics.

a Optimizing delivery routes

b Detecting diseases from X-ray images

c Detecting pests and diseases

d Automating product quality inspection

⚙ Think as an Engineer — Investigate & Decide

You have just discussed agriculture in the Think It Through note. Now investigate a different industry and decide.

1 · Investigate. Choose one industry from the lesson other than agriculture — healthcare, manufacturing, or logistics. Find one real, documented example of AI being used in it, and record the source you used.

2 · Weigh it up. Note one benefit and one caution of using AI in that industry, using the ideas of what AI is good at and what needs caution.

3 · Decide. Should this industry use AI for this task on its own, or with a human checking the result? Recommend one approach and give two reasons.

Stuck? AI is good at finding patterns quickly; but a human is needed where a wrong result could harm someone or involve responsibility.

🌍 In a New Context

A school library wants to use AI to recommend books to students based on what they have borrowed before. Using what you learned about recommendation systems and what AI is good at, explain how the AI would work, and identify one caution the school should consider.

💡 THINK IT THROUGH

In agriculture, give one realistic advantage of introducing AI and one concern that should be considered before introduction, and briefly explain the reason for each.

❓ Lesson Question — Answered

AI is used widely in daily life — in recommendation systems, voice assistants, machine translation, and face recognition — and in industry, across fields such as healthcare, agriculture, manufacturing, and logistics. AI is good at finding and classifying patterns in data, recognizing and generating images, audio, and text, and making probabilistic predictions from data. However, it needs caution where ethical judgments, personal information and privacy, and final decision-making are involved, and where training data is biased. So AI is a powerful tool for pattern-based tasks, but human judgment is still needed for responsibility, ethics, and fairness.

Exercise

1 Cover the Point! section with a red plastic sheet and quiz yourself in your notebook in numerical order.

2 Answer the following questions.

(1) Select all options that AI excels at.

A Ethical judgments

B Finding features in images and text and classifying them

C Judgments involving personal information

D Probabilistic reasoning and prediction based on data

E Decision-making that bears responsibility for results

F Recognition and generation of images, audio, and text

(2) From the following options (A–D), choose the one that is NOT an appropriate description of services that use AI.

A A recommendation system predicts preferences from past behavior data and displays recommendations.

B A voice assistant recognizes a voice, understands commands, and executes them.

C Machine translation is a system in which humans manually translate foreign languages.

D Face recognition automatically detects and identifies human faces in photographs.

3 Read the following passage and answer each question.

AI is widely used in our daily lives. For example, recommendations of videos that match one's preferences on a video site come from a (a) system. Also, AI

⭐ REFLECT & CHALLENGE

Reflect: Which use of AI in this lesson surprised you most? Where in your own life do you think AI is making decisions about you? **Challenge:** Choose one task where AI needs caution (ethics, privacy, or final decisions). Suggest one rule that would make using AI for that task safer.

technology is used in (b), with which one can speak to a smartphone to operate it, and in (c), which converts foreign-language text into one's native language.

1. Fill in the blanks (a)–(c).

(2) From the following (a–d), choose all that require caution when using AI.

a High-speed processing of large amounts of data

b Ethical judgments

c Judgments involving personal information or privacy

d Recognition of images and audio

⭐ Key takeaway

Remember:
AI is good at finding and classifying patterns in data — recommendations, voice, translation, face recognition, and many industry tasks — but human judgment is still needed where ethics, privacy, and responsibility are involved.